



BHUMIKA RAJ

Transportation Planner



Experience WSP

2 years and 10 months

EDUCATION

Undergrad in Civil Engineering (Major in Construction Management)
CEPT University, India

Master's in City and Regional Planning
Rutgers University, New Jersey

Areas of Practice

Transit & Rail Service Planning

Statewide / Regional Planning (TIP / STIP)

Grant & Program Support

Workflow Automation (Microsoft Power Platform)

Project Management

Transit Wayfinding

Data Analysis

Technical Skills

ArcGIS

Remix

Stata

Power BI

Power Automate

MS Project

CAREER SUMMARY

Ms. Raj is a transportation planner with undergraduate in Civil Engineering. She has experience on Real estate projects, project, and stakeholder coordination of various projects nationally and internationally. She interned with Gateway Program including projects like Penn station expansion, Hudson Tunnel and Harrison Track project. Further she also interned on Mumbai Metro line 3, one of the first underground metro project for the state of Maharashtra, India. She assisted with preparation of internal real estate Strategic plan for Amtrak's Gateway Program, property acquisition for Penn station Expansion project and PMP for Hudson Tunnel. She has also worked in areas of active transportation projects, micro transit services, and mobility as a service (Maas) concept for Jersey City. Currently Ms Raj is an embedded consultant at Maryland Transit Administration and works with WSP supporting various clients across Mid-Atlantic.

WSP EXPERIENCE

MTA- Maryland Transit Administration, Office of Statewide Planning- Part-time Embedded Consultant (ongoing)

Ms. Raj serves as Transit Planner in the **Office of Statewide Planning at MTA**, specializing in regional planning initiatives and the MARC Commuter Rail Line in Maryland.

Regional Planning Tasks (Statewide): Works directly with Chief of Strategic Planning at MTA Maryland Office of Statewide Planning to perform TIP, STIP Amendments, Clearing house reviews, Correspondence responses on behalf of Governor's office and other tasks.

Provides key support to MTA's Office of Statewide Planning for updating their briefing books and information on projects to be included in **CTP- Consolidated Transportation Plan Tour** [MDOT Consolidate Transportation Plan](#) which are submitted to governor and state transportations secretary, and used in public meetings and tours.

Prepared Service Proposals Evaluating route for Commuter Bus Service Proposals for Maryland, which were submitted to MTA transportation secretary for next steps.

Silver Spring Turnback Feasibility Study (\$1 Mil): PM for the project which is a crucial enabling project of MARC Growth and Transformation Plan (G&T) [MARC Growth and Transformation Plan](#), a key plan to improve MARC service in State of Maryland

Silver Spring turnback 15% Design (\$3Mil) PM for the project which is a crucial enabling project of MARC G&T, a key plan to improve MARC service in State of Maryland. [MARC G&T Initial Investment Projects](#)

MARC Wayfinding Manual and Signage Report: Project Manager for the Wayfinding initiative in state of Maryland across three MARC (42 Stations)- Maryland Area Rail Commuter lines, including development of GIS Survey forms, GIS Inventory for data storage, comprehensive wayfinding booklets, and MARC signage manuals. Helping clients do it in-house has helped save thousands of dollars and build institutional capability.



Veriato (Beginner level)

Professional Affiliations

Member YPT-Young
Professionals in
Transportation

(Board Member- Director of
Inter-Chapter Affairs and
Membership YPT-2024)

Fed State Partnership Grant: Coordinated and led the efforts of the FSP Grant for MTA and City of Rockville for Rockville and Brunswick station pedestrian improvement directly reporting to Director of Statewide Planning at MTA.

Public Comment Analysis Tool: Leading the Public Comment Tool App, for large scale projects which would automate the process of extraction, response, documentation, and sentiment analysis for comments received by public instead of relying on third party tools saving costs and managing data's confidentiality.

Microsoft Power Platform Automation: Leads automation of the clearinghouse process and Comprehensive Plan reviews using Microsoft Power Automate tool to streamline operational workflows. Reducing time significantly by 70%.

FEMA- Federal Emergency Management Association

Presented the **AI Ordinance tool** to FEMA (700 Mil JV) Deputy Branch chief for Mid-Atlantic region-this is a first of a kind tool to automate the process of hundreds of ordinances received annually and would help **reduce time by 50%** once deployed.

Leading the effort to submit the design documentation scope for the Project, next step is execution of the design document.

SJTPO-Southern Jersey Transportation Planning Organization, TIP Development (New Jersey)

Ms. Raj helped update the TIP-Transportation Improvement Plan for SJTPO, prepared a TIP Peer comparison memo for analysis across four different agencies, and created a County one-pager for public meetings.

MTA Unhoused Study Project (MTA, Baltimore)

Ms. Raj worked with MTA in preparation for an internal report addressing homelessness within MTA Assets. Tasks included Data Collection and Analysis, Literature review, Coordination with internal MTA departments and preparing a comprehensive report as deliverable.

East Side Transportation Plan (City of Fort Worth, Fort Worth Texas)

Ms. Raj worked with the City of Fort Worth on their east side transportation plan for providing a Gentrification Impact Study. The study is to evaluate the potential of high-capacity transit, improvements for other transit and transportation facilities in the area, and to support economic development and the City's land use policies. The study aims to create pedestrian solutions and build consensus for future improvements.

Safety Risk Assessment Plan, Washington Metropolitan Area Transit Authority (WMATA), Washington DC, Transportation Planner (2023)

Ms. Raj served as a Transportation Planner and Data Analyst for a safety risk assessment plan at WMATA, contributing to the development of the SMS Safety Risk Management System. The WSP Team worked with subconsultants to deliver the project. The project received the **ACEC Metro Washington Award 2023**.

PREVIOUS EXPERIENCE:

- **Amtrak, Gateway Program Management Intern Manhattan, New York**
[Amtrak Gateway Program](#) Ms. Raj collected data and coordinated with the senior director of portfolio to conduct the 8-hour workshop on the project delivery methodology of the Harrison Fourth Track project in New Jersey with project and program managers. Documented the methodology for the team that identified the suitable design-bid-build contract for the project to enter the next engineering phase. The process also provided credibility for justifying the selected delivery type applying for the FTA 's CRISI funding program.

Calculated the transfer taxes for more than one hundred properties needed for surface easements of the **Penn Station Expansion Project** by leveraging Zola to map parcels and lots in New York's Midtown Manhattan, which was then submitted to the New York City Council for decision-making.

Attended the weekly meetings and prepared notes for **real estate strategic documents** with the senior director of portfolio for property acquisitions strategy and Empire State Development for a general project plan for the redevelopment of Midtown Manhattan in conjunction with the Penn Station Expansion Project.

Coordinated with the gateway program manager for daily tasks; memorandum writing; and weekly meetings with subconsultants and stakeholders. Prepared (RACI) responsibility, accountability, consultancy, and information matrix for all active projects and assisted in SharePoint design and management.

Formulated lessons learned logs for all six projects under the Gateway Program to identify major risks to avoid in the different project phases and to celebrate success scenarios deriving insights from the lessons learned report for the Moynihan Train Hall Construction Project.

- **Leveraging Value Capture Techniques in Freights and Ports, New York, and New Jersey: (Pilot Bus Study by Texas A&M)** Ms. Raj conducted the literature review and provided suitable recommendations building upon the Freight NYC Plan by NYCEDC. Analyzed the case study of Texas (NBTA) and recommended alternate delivery model for NYCDOT to capture the value of public transportation– Using SELECT Bus service/ Public transportation for last mile delivery of goods from warehouses to reduce congestion in Midtown Manhattan.
- **Mobility Hub Identification along Hudson Bergen Light Rail- Jersey City, New Jersey** Ms. Raj led the research and proposal model for implementing mobility as a service for the Jersey City Council. Reviewed the existing report and filled in the gaps using national and global case studies for micro-transit services and funding mechanism models to improve its efficiency and scalability and make it more equitable. She used system thinking to create a diagram identifying leverage points, impacted areas, and ideas, which she presented to the Jersey City Council.

- **Civil Engineering Intern, Mumbai Metro Line 3, Maharashtra, India**
[Mumbai Metro Line 3](#) Ms. Raj learned on-site and administrative aspects of the functioning of the metro from the client as well as the contractor's perspective and prepared a detailed report of 150 pages as part of the curriculum. Prepared in-depth feasibility report of the project's corridor and alignment study. She prepared reports on resource scheduling, value engineering, cost-benefit analysis, quantity estimation, contract management, project delay, factor analysis, environmental impact assessment, and socioeconomic report study.

Visited the construction site every week as part of the curriculum report. She evaluated construction methodologies, including New Austrian Tunneling Method, cut and cover, concreting, reinforcement, and formwork activities in terms of design quality, safety, productivity, cost, and time cycle analysis.